

xv6 memory, schedule improvement agent

Team OS_JPLKJ

2022270638_조현성

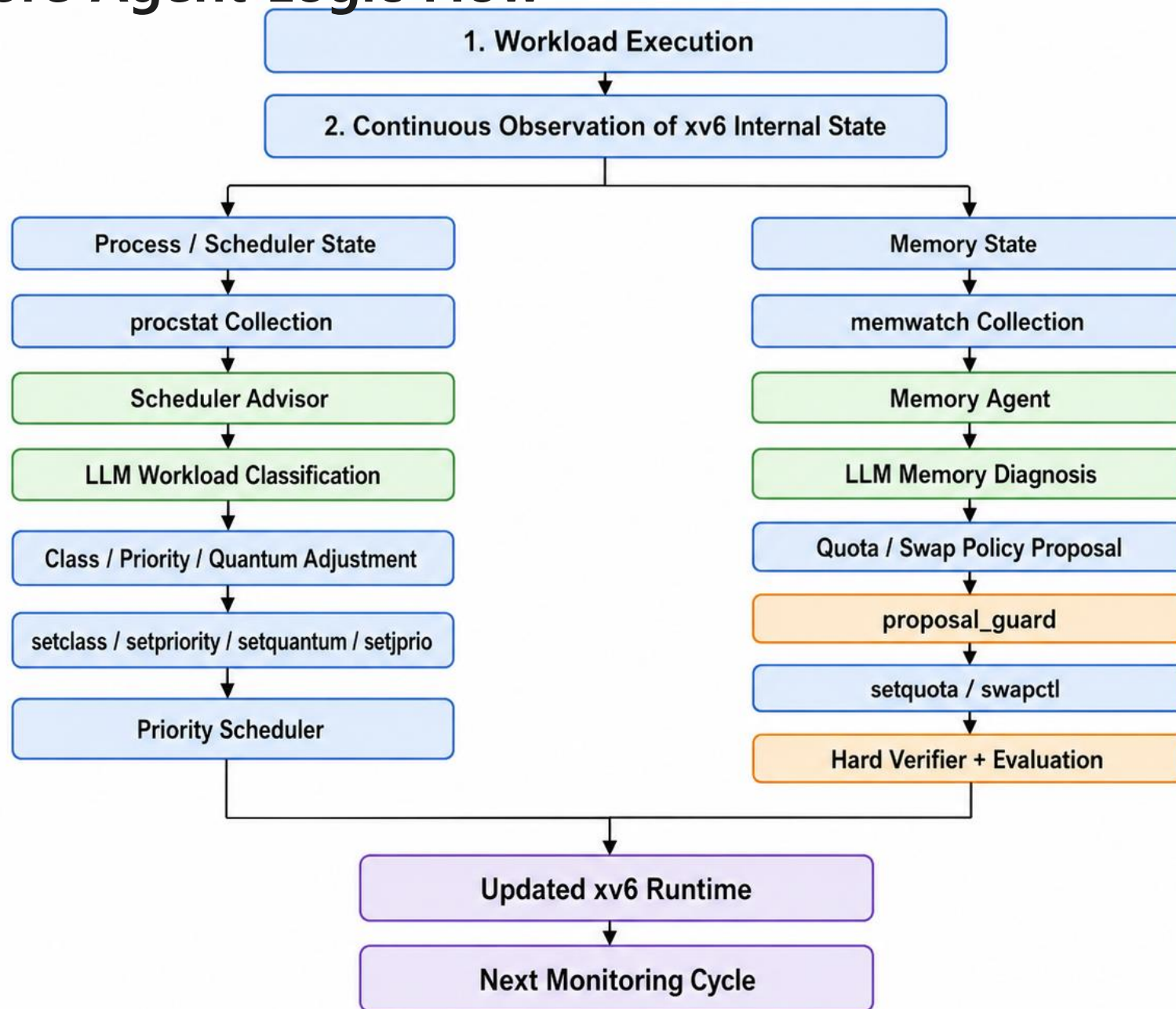
2022270654_박규태

2021270603_이찬주

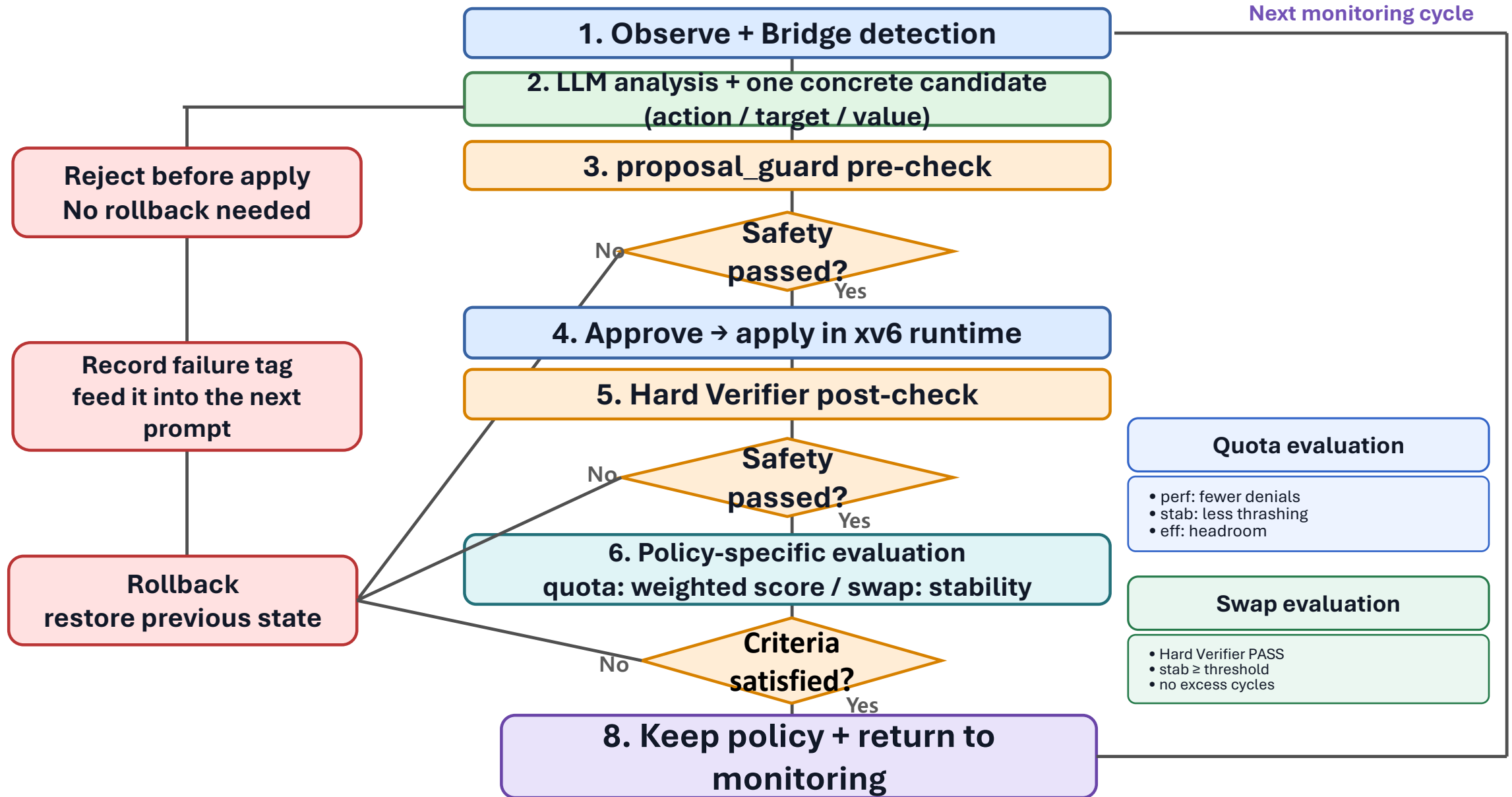
2022270656_김경선

2022270620_조승찬

Integrated Core Agent Logic Flow



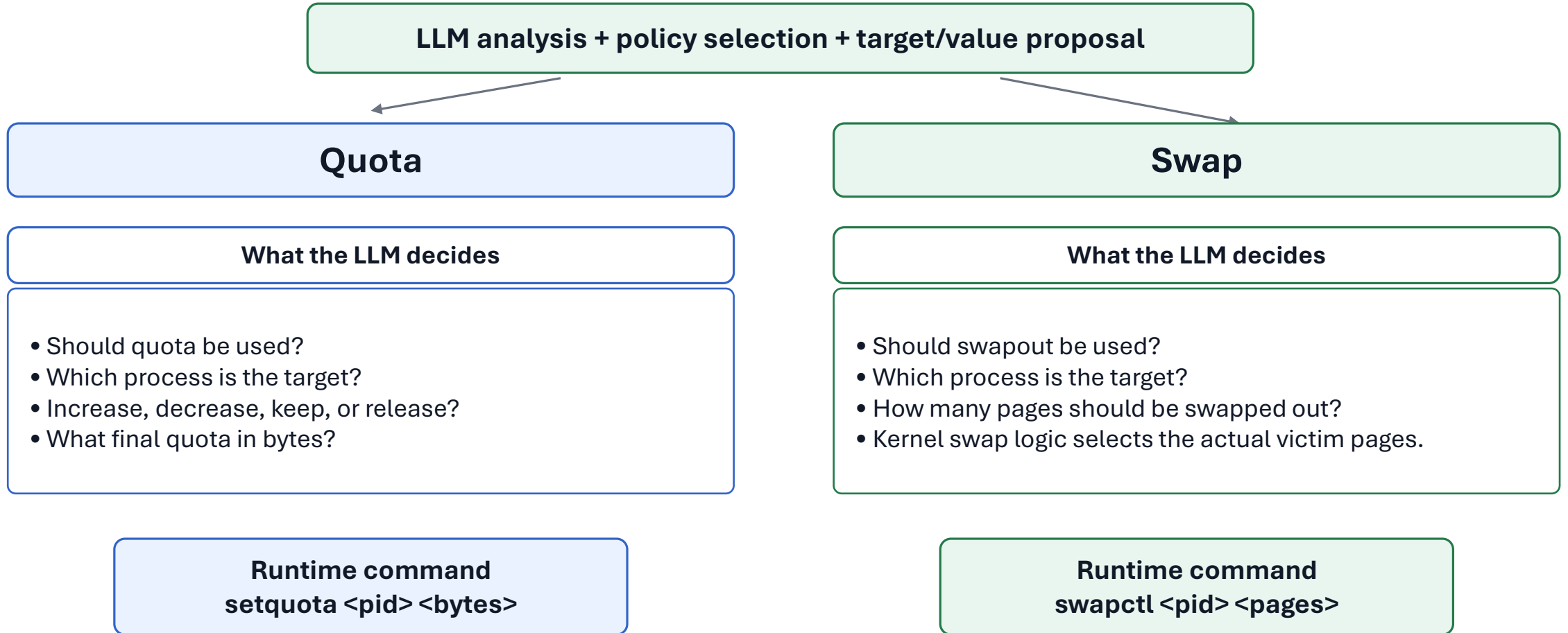
Core Agent Logic Flow/memory



Key point: the LLM proposes; guard, verifier, and metrics decide whether to keep it.

Core Policy Modules: Quota and Swap

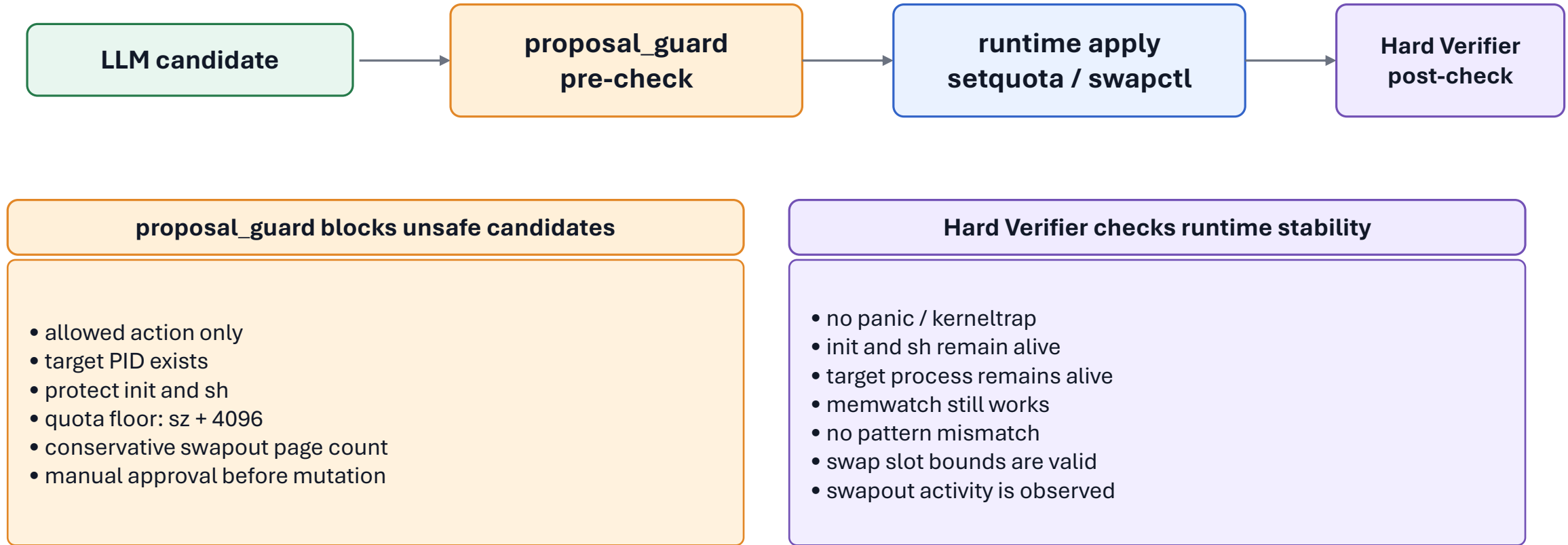
The LLM selects the policy and proposes the concrete value to apply.



Each separates the policy into a modular form and effectively reduces meaningless loops in the form of llm proposing fine-tuning.

Dual Safety Checks: Before and After Runtime Apply

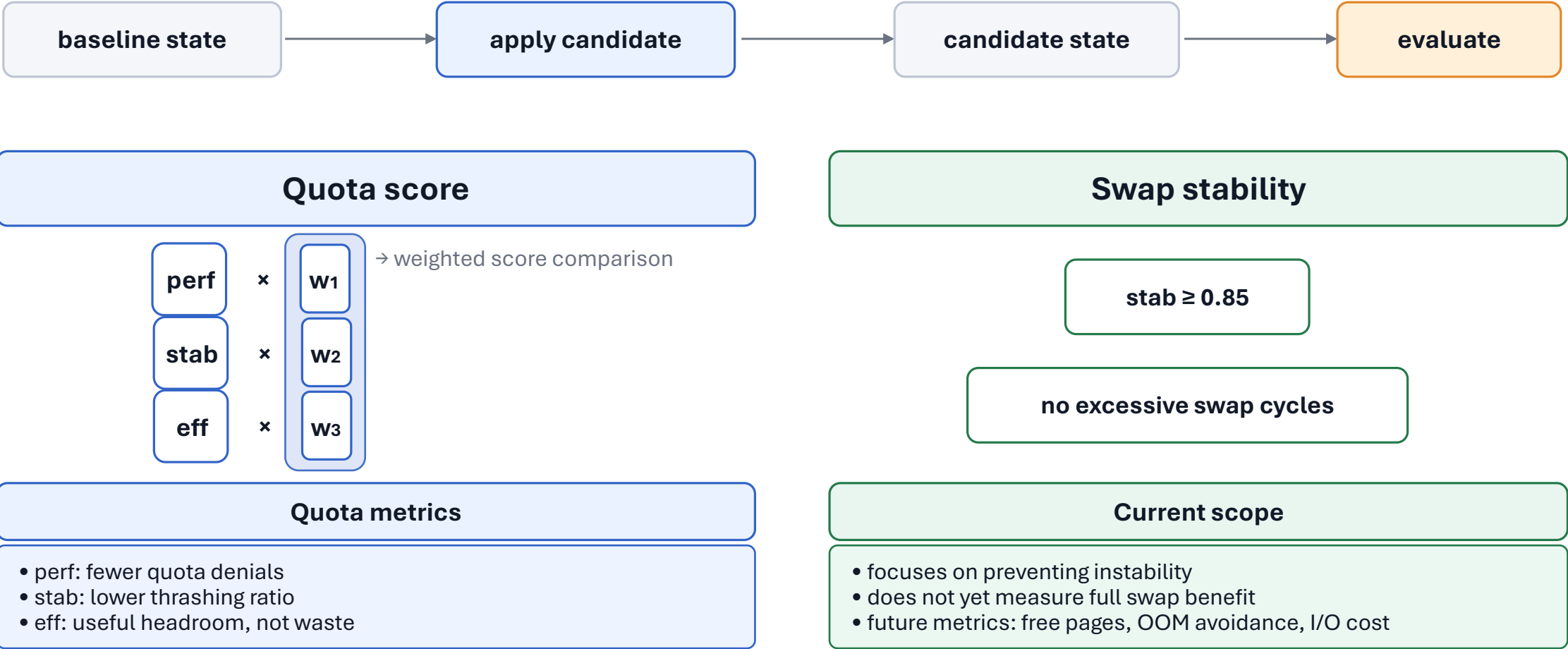
Two different checks reduce unsafe loops and catch different failure modes.



Guard "blocks dangerous proposals" and Verifier "check kernel stability after application"

Evaluation and Rollback: Quota Uses Score, Swap Uses Stability

The LLM does not score policies; Python code evaluates fixed metrics.



Each calculation and standard are set aside to increase reliability

Results/quota

eng

```
===== QUOTA SCORE RESULT =====
1:[Agent] Preparing the QEMU environment... Rebuilding the kernel if necessary.
2:[Agent] QEMU environment ready
5:[Observation 01] Anomaly Detected
8:LLM Diagnosis      : Quota-pressure scenario for executor validation
9:Selected Policy    : Increase quota
12:Applied Command   : setquota 4 155648
13:Verification      : PASS
14:Evaluation        : score 79.7 →81.7
15:Final Decision    : Keep applied policy
16:Next Step         : Return to monitoring loop
19:[Observation 02] Anomaly Detected
22:LLM Diagnosis     : Quota-pressure scenario for executor validation
23:Selected Policy    : Increase quota
26:Applied Command   : setquota 4 159744
27:Verification      : PASS
28:Evaluation        : score 81.7 →83.5
29:Final Decision    : Keep applied policy
30:Next Step         : Return to monitoring loop
33:[Observation 03] Anomaly Detected
36:LLM Diagnosis     : Quota-pressure scenario for executor validation
37:Selected Policy    : Increase quota
40:Applied Command   : setquota 4 163840
41:Verification      : PASS
42:Evaluation        : score 83.5 →85.2
43:Final Decision    : Keep applied policy
44:Next Step         : Return to monitoring loop
49:Final Result      : COMPLETED
51:QEMU Restarts     : 0
```

kor

```
===== QUOTA SCORE RESULT =====
5:[관찰 01] 이상 감지
8:LLM 분석          : 시험용 quota 압박 상황
9:선택 정책          : quota 증가
12:적용 명령         : setquota 4 155648
13:검증 결과         : PASS
14:평가 결과         : score 79.7 →81.7
15:최종 판단         : 정책 적용 유지
16:후속 동작         : 감시 루프로 복귀
19:[관찰 02] 이상 감지
22:LLM 분석          : 시험용 quota 압박 상황
23:선택 정책          : quota 증가
26:적용 명령         : setquota 4 159744
27:검증 결과         : PASS
28:평가 결과         : score 81.7 →83.5
29:최종 판단         : 정책 적용 유지
30:후속 동작         : 감시 루프로 복귀
33:[관찰 03] 이상 감지
36:LLM 분석          : 시험용 quota 압박 상황
37:선택 정책          : quota 증가
40:적용 명령         : setquota 4 163840
41:검증 결과         : PASS
42:평가 결과         : score 83.5 →85.2
43:최종 판단         : 정책 적용 유지
44:후속 동작         : 감시 루프로 복귀
49:최종 결과         : COMPLETED
51:QEMU 재시작       : 0
```

Results/swap

eng

```
===== SWAP EXECUTOR RESULT =====
1:[Agent] Preparing the QEMU environment... Rebuilding the kernel if necessary.
2:[Agent] QEMU environment ready
5:[Observation 01] Anomaly Detected
8:LLM Diagnosis      : RAM-pressure scenario for swap executor validation
9:Selected Policy    : swapout
12:Applied Command   : swapctl 4 1
13:Verification      : PASS
14:Evaluation        : stab=1.000, cycles=0
15:Final Decision    : Keep applied policy
16:Next Step         : Return to monitoring loop
19:[Observation 02] Anomaly Detected
22:LLM Diagnosis     : RAM-pressure scenario for swap executor validation
23:Selected Policy   : swapout
26:Applied Command   : swapctl 4 1
27:Verification      : PASS
28:Evaluation        : stab=1.000, cycles=0
29:Final Decision    : Keep applied policy
30:Next Step         : Return to monitoring loop
33:[Observation 03] Anomaly Detected
36:LLM Diagnosis     : RAM-pressure scenario for swap executor validation
37:Selected Policy   : swapout
40:Applied Command   : swapctl 4 1
41:Verification      : PASS
42:Evaluation        : stab=1.000, cycles=0
43:Final Decision    : Keep applied policy
44:Next Step         : Return to monitoring loop
49:Final Result      : COMPLETED
51:QEMU Restarts     : 0
```

kor

```
===== SWAP EXECUTOR RESULT =====
5:[관찰 01] 이상 감지
8:LLM 분석          : 시험용 RAM 압박 상황
9:선택 정책          : swapout
12:적용 명령         : swapctl 4 1
13:검증 결과         : PASS
14:평가 결과         : stab=1.000, cycles=0
15:최종 판단         : 정책 적용 유지
16:후속 동작         : 감시 루프로 복귀
19:[관찰 02] 이상 감지
22:LLM 분석          : 시험용 RAM 압박 상황
23:선택 정책          : swapout
26:적용 명령         : swapctl 4 1
27:검증 결과         : PASS
28:평가 결과         : stab=1.000, cycles=0
29:최종 판단         : 정책 적용 유지
30:후속 동작         : 감시 루프로 복귀
33:[관찰 03] 이상 감지
36:LLM 분석          : 시험용 RAM 압박 상황
37:선택 정책          : swapout
40:적용 명령         : swapctl 4 1
41:검증 결과         : PASS
42:평가 결과         : stab=1.000, cycles=0
43:최종 판단         : 정책 적용 유지
44:후속 동작         : 감시 루프로 복귀
49:최종 결과         : COMPLETED
51:QEMU 재시작       : 0
```

LLM Scheduler Demonstration

1. Immediate name-based classification with automatic priority and quantum adjustment



```
userkks@DESKTOP-2V387VK:~/projects/OS_JPLKJ/xv6-process$ python3 tools/bridge.py --work
load spin,cc --duration 120
[bridge] spawning: qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m
128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=n
one,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0 -chardev
socket,path=/tmp/xv6advisor.sock,server=on,wait=off,id=br0 -device virtio-serial-device
,bus=virtio-mmio-bus.1 -device virtconsole,chardev=br0
[bridge] advd launched; advisor channel connected on /tmp/xv6advisor.sock; classify loo
p
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 4266ms
[bridge] (pid=3 name=sh NORMAL -> INTERACTIVE (run=1 sleep=2) via LLM
[bridge] (pid=7 name=cc BATCH -> CPU_BOUND (run=2 sleep=0) via LLM
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 4449ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 4156ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 4949ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 4947ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 5301ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 5184ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 5159ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 5271ms
[bridge] guest: @00K setcls pid=3 cls=0 prio=4
[bridge] guest: @00K setcls pid=7 cls=3 prio=12
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 4703ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 5243ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 5364ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 5339ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 5687ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 6091ms
[bridge] pid=7 name=cc CPU_BOUND -> BATCH (run=71 sleep=0) via LLM
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 5649ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 6290ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 6508ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 6860ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 6876ms
[bridge] LLM(qwen2.5:3b) classified 3 proc(s) in 8756ms
[bridge] LLM unavailable/invalid (TimeoutError: timed out); falling back to heuristic
[bridge] pid=5 name=spin NORMAL -> CPU_BOUND (run=108 sleep=0) via heuristic
[bridge] duration reached; shutting down QEMU
```

2. Pattern-based learning accumulated by process name after termination



```
userkks@DESKTOP-2V387VK:~/projects/OS_JPLKJ/xv6-process$ make qemu
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nogr
aphic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x
0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0 -chardev socket,path=/tmp/xv
6advisor.sock,server=on,wait=off,id=br0 -device virtio-serial-device,bus=virtio-mmio-bu
s.1 -device virtconsole,chardev=br0
```

xv6 kernel is booting

hart 2 starting

hart 1 starting

init: starting sh

\$ spin &

\$ spin &

\$ spin &

\$

1 sleep init prio=10 class=2 q=1 run=0 rdy=1 slp=329 cs=28

3 sleep sh prio=10 class=2 q=1 run=0 rdy=0 slp=329 cs=22

5 run spin prio=10 class=2 q=1 run=247 rdy=0 slp=0 cs=209

7 run spin prio=10 class=2 q=1 run=244 rdy=3 slp=0 cs=223

9 run spin prio=10 class=2 q=1 run=230 rdy=5 slp=0 cs=200

\$ kill 5

\$ kill 7

\$ kill 9

\$ priors

learned priors: 1 entries

name / class / samples / avg_run / avg_sleep

spin -> CPU_BOUND samples=3 run=574 sleep=0

\$

3. Job-group management with fork-bomb detection and mitigation



```
userkks@DESKTOP-2V387VK:~/projects/OS_JPLKJ/xv6-process$ python3 tools/bridge.py --work
load ftree --duration 60
[bridge] spawning: qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -
m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=n
one,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0 -chardev
socket,path=/tmp/xv6advisor.sock,server=on,wait=off,id=br0 -device virtio-serial-device
,bus=virtio-mmio-bus.1 -device virtconsole,chardev=br0
[bridge] advd launched; advisor channel connected on /tmp/xv6advisor.sock; classify loo
p
[bridge] (group 5: 13 proc(s) ['ftree'] run~0 life>=0 -> bomb via LLM
[bridge] (PRE-EMPT fork-bomb: setjprio 5 19)
[bridge] LLM unavailable/invalid (JSONDecodeError: Un
[bridge] LLM unavailable/invalid (JSONDecodeError: Un
[bridge] pid=3 name=sh NORMAL -> INTERACTIVE (run=0 sleep=7) via heuristic
[bridge] LLM unavailable/invalid (TimeoutError: timed out); falling back to heuristic
[bridge] duration reached; shutting down QEMU
```

4. Kernel panic timeline analysis with LLM-based natural language diagnosis



```
==== kernel panic timeline ====
@@PANIC reason="oom: kalloc returned 0" tick=2 events=14
@@EV tick=1 type=EXIT pid=2 name=priors d1=0 d2=0 msg=
@@EV tick=2 type=NOTE pid=4 name=crashme d1=0 d2=0 msg=build: pipeline start
@@EV tick=2 type=EXIT pid=5 name=crashme d1=0 d2=0 msg=
@@EV tick=2 type=FORKFAIL pid=4 name=crashme d1=4 d2=16 msg=cap
@@EV tick=2 type=FORKFAIL pid=4 name=crashme d1=4 d2=16 msg=cap
@@EV tick=2 type=FORKFAIL pid=4 name=crashme d1=4 d2=16 msg=cap
@@EV tick=2 type=FORKFAIL pid=4 name=crashme d1=4 d2=16 msg=cap
@@EV tick=2 type=FORKFAIL pid=4 name=crashme d1=4 d2=16 msg=cap
@@EV tick=2 type=FORKFAIL pid=4 name=crashme d1=4 d2=16 msg=cap
@@EV tick=2 type=FORKFAIL pid=4 name=crashme d1=4 d2=16 msg=cap
@@EV tick=2 type=FORKFAIL pid=4 name=crashme d1=4 d2=16 msg=cap
@@EV tick=2 type=FORKFAIL pid=4 name=crashme d1=4 d2=16 msg=cap
@@EV tick=2 type=NOTE pid=4 name=crashme d1=0 d2=0 msg=fork rate exploding
@@PANIC_END

(==== diagnosis (LLM qwen2.5:3b) ====)
The sequence of events led to the panic starting with a job-group 4 process named `cras
hme` repeatedly hitting FORKFAIL, each time refusing its excess forks due to the per-jo
b cap (job-group 4 already held 16 live processes). The process was continuously spawni
ng new instances at tick 2. Each instance hit the FORKFAIL condition multiple times, wi
th messages indicating that the fork rate was exploding. This pattern of repeated FORKF
AILS from a single process is indicative of a fork-bomb scenario being contained by the
anti-fork-bomb defense.

The most likely root cause for the kernel panic at tick 2 is an out-of-memory (OOM) con
dition triggered by this excessive forking, despite the cap on live processes. The pani
c reason "oom: kalloc returned 0" confirms that the kernel encountered a situation wher
e memory allocation failed, leading to the panic. This indicates that while the FORKFAI
L defense was engaged and working as designed, it did not prevent the system from runni
ng out of available memory due to the high rate of forking by the `crashme` process.
```